American Computer Science League

2025-2026 ● Contest 1: Short Problems Solutions ● Classroom Division

1. Computer Number Systems

To convert from hexadecimal to octal, it is easiest to use 4 bits for hexadecimal and then 3 bits for octal.

$52326_{16} = 0101\ 0010\ 0011\ 0010\ 0110_2$
$= 01\ 010\ 010\ 001\ 100\ 100\ 110_2$
$= 1\quad 2\quad 2\quad 1\quad 4\quad 4\quad 6_8$
$= 1221446_8$

B. 1221446

2. Computer Number Systems

$(36_8 + 74_8) / 2$
$= 132_8 / 2$
$= (1*8^2 + 3*8 + 2) / 2$
$= (64 + 24 + 2) / 2$
$= 90_{10} / 2$
$= 45_{10}$

C. 45

3. Recursive Functions

$f(1) = 2$
$f(2) = 2 * f(2-1) + 1 = 2 * f(1) + 1 = 2 * 2 + 1 = 5$
$f(3) = 2 * f(3-1) + 1 = 2 * f(2) + 1 = 2 * 5 + 1 = 11$
$f(4) = 2 * f(4-1) + 1 = 2 * f(3) + 1 = 2 * 11 + 1 = 23$
$f(5) = 2 * f(5-1) + 1 = 2 * f(4) + 1 = 2 * 23 + 1 = 47$
$f(6) = 2 * f(6-1) + 1 = 2 * f(5) + 1 = 2 * 47 + 1 = 95$
Now substitute backwards.
The sequence is: 2, 5, 11, 23, 47, 95, …
Each term can be represented by $3 * 2^{n-1} - 1$.

D. 95

4. Recursive Function

$f(16) = f(16-3) + 16 = f(13) + 16 = 73 + 16 = 89$
$f(13) = f(13-3) + 13 = f(10) + 13 = 60 + 13 = 73$
$f(10) = 2 * f(10-2) = 2 * f(8) = 2 * 30 = 60$
$f(8) = 2 * f(8-2) = 2 * f(6) = 2 * 15 = 30$
$f(6) = 2 * 6 + 3 = 15$
Now substitute backwards.

E. 89

5. Computer Number Systems

Evaluate in octal:

$2025_8 - 202_8 + 20_8 / 2_8 = 2025_8 - 202_8 + 10_8$
$= 1623_8 + 10_8$
$= 1633_8$

C. 1633

6. Computer Number Systems

The first 10 Fibonacci numbers when converted to binary are:

0 ⇒ 0 1 ⇒ 1 1 ⇒ 1
2 ⇒ 10 3 ⇒ 11 5 ⇒ 101
8 ⇒ 1000 13 ⇒ 1101 21 ⇒ 10101
34 ⇒ 100010

Therefore there are 16 1s in the binary notation.

C. 16

7. Recursive Functions

A 16x16 square becomes an 8x8 square, then a 4x4, then a 2x2, then a 1x1, and finally a 0.5x0.5. Therefore, its perimeter is 4*0.5 = 2.

C. 2

8. Recursive Functions

$f(16, 12) = f(16 - 2, 12 + 1) + 12 = f(14, 13) + 12 = 7$
$f(14, 13) = f(14 - 2, 13 + 1) + 13 = f(12, 14) + 13 = -5$
$f(12, 14) = f(12 + 1, 14) - 12 = f(13, 14) - 12 = -6 - 12 = -18$
$f(13, 14) = f(13 + 1, 14) - 13 = f(14, 14) - 13 = 7 - 13 = -6$
$f(14, 14) = 14/2 = 7$ Now substitute backwards.
Also, $f(16, 12) = 7 - 13 - 12 + 13 + 12 = 7$

B. 7

9. What Does This Program Do?

```
input a, b, c
if c == 0 then
    c = sqrt(a^2 + b^2)
else
    if b == 0 then
        b = sqrt(c^2 - a^2)
    else
        a = sqrt(c^2 - b^2)
    end if
end if
output a,b,c
```

The program finds the missing value in a right triangle so in each case, the value that is 0 is replaced with the correct value. This problem is in honor of Pythagorean Theorem Day on 7/24/25.

D. All of the above

10. What Does This Program Do?

```
a = 13
b = 21
n = 8
while sqrt(b)!=int(sqrt(b))
    n = n + 1
    c = a + b
    a = b
    b = c
end while
output n
```

The loop will stop when 144 is a perfect square. It is the 12th Fibonacci number in the sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 since 21 is the 8th one.

A. 12